2025A Q14

Wednesday, April 22, 2026 1:24 PM

Checklist

- [ ] Basic Programming
  integer , ← variable
- [ ] Array Data Structure
  2D
- [x] for loop | nested for loop
- [x] Sum of an array OK

array

4	2	1	6
1	2	3	4

index

sum $\boxed{0}$ why 0

0 – ออกจาก

Forumla ⟸ sum + array[i]

7 + 6

for i = 1 array_data – 13

i = 2 4

i = 3 2

i = 4 1

6

sum ← 13

i = 1 to i = 4

product $\boxed{1}$

Formula

prod ← prod * array[i]

array_data

i = 1

i = 2

i = 3

i = 4

SUM-ARRAY(A, n)

1 sum ← 0

2 for i = 1 to n

3 sum ← sum + ~~A[i]~~ m[i,i]

4 return sum

i=1 sum + arry[1] → sum

0 + 4 → 4

i=2 sum + arry(2)

4 + 2

6

i=3 sum + arry(3)

6 + 1

7

sum = 7

i=4 sum + arry[4]

7 + 6

13

i=1 i=2 i=3

dia 1 1,1 2,2 3,3

dia 2 1,3 2,2 3,1

4	9	2	→ 15
3	5	7	→ 15
8	1	6	→ 15
↓ 15	↓ 15	↓ 15	↘ 15

j = 1 2 3

i = 1 2 3

j = 1 2 3

i =

row sum = [15, 15, 15]

1 2 3

colsum = [15, 15, 15]

1 2 3

k = 1 → rowsum[k] = rowsum[1] = 15 — 15

2 = 15 — 15

3 = 15 — 15

k = 2

3

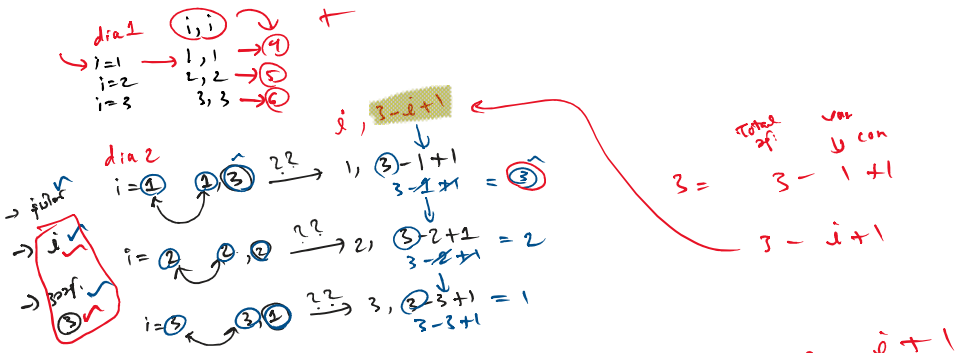
dia 1 1,1 2,2 3,3

dia 2 1,3 2,2 3,1

i,i

i=1 → 1,1 → 4

+



Why OR but not AND

row sum[k] = 15
col sum [k] = 15
dial sum = 20

- `rowSum[k] ≠ colSum[k]` → `15 ≠ 15` → **false**
- `rowSum[k] ≠ dia1Sum` → `15 ≠ 20` → **true**

if (~) → ဖေါ်ပြထားသည့်

return false → ဖေါ်ပြထားသည်

OR

false OR ture → true

if (true) → ဖေါ်ပြထားတာ မှန်ရင်ပေါ် မှန်ပါတယ်

→ return false → ဖေါ်ပြထားသည်

AND

false AND true → false

if (flase) → ဖေါ်ပြထားတာ မမှန်ရင်ပေါ် ကျော်ရမယ်

return false